BAECE203 – ANALOG ELECTRONICS

DIGITAL ASSIGNMENT – REVIEW 0

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TITLE OF THE PROJECT

“Statistical and Post-Layout Characterization of a Gm-C Continuous-Time Filter in the Open-Source SKY130 PDK: A Monte Carlo, PVT-Corner, and Cross-Validation Study”

PROBLEM STATEMENT

Gm-C filters are chosen in low-power analog front-ends specifically because their cutoff frequency and quality factor can be tuned electronically via bias current — but this same current-dependence makes their performance directly vulnerable to random transistor-to-transistor mismatch, an effect that grows more significant as devices are scaled down for area and power efficiency. Standard filter design practice verifies only the nominal-case frequency response against hand-calculated design equations, which reveals nothing about how far a real, manufactured population of the same filter would actually deviate from that target — a gap of direct consequence once a filter is deployed inside a resolution- or power-constrained system such as an ECG front-end or an ADC anti-aliasing stage.

This gap is compounded in the specific context of open-source silicon design. The SkyWater SKY130 process is increasingly used by students, researchers, and the open-hardware community as a free alternative to commercial process design kits, yet its built-in statistical mismatch models — the very thing a Monte Carlo characterization would need to trust — have no independently measured, publicly available silicon dataset against which their accuracy can be checked. A designer using SKY130 today has no way to know whether a statistical result obtained from its models reflects real manufacturing variation or an artifact of an unverified model file.

This project therefore addresses two linked problems: first, quantifying how much a representative Gm-C filter's cutoff frequency, quality factor, passband ripple, and power consumption actually vary under realistic device mismatch and PVT conditions in SKY130; and second, establishing whether the statistical models producing those numbers can be trusted at all, given the complete absence of measured silicon data to check them against — answered here through an independent analytical cross-validation against the classical Pelgrom mismatch law rather than by assumption.

ABSTRACT

Continuous-time Gm-C filters provide electronically tunable, low-power frequency response and are widely used in biomedical acquisition, sensor interfaces, and channel-select front-ends, yet their sensitivity to random device mismatch remains unquantified in open-source silicon design kits, and no independently measured mismatch dataset exists publicly for the SkyWater SKY130 process to validate its built-in statistical models. This work designs a second-order Gm-C low-pass filter in SKY130 — two transconductor-capacitor integrator stages built from differential-pair OTAs — in Xschem/ngspice, and characterizes it through 500-run Monte Carlo mismatch simulation, a five-corner process-voltage-temperature sweep, and an independent analytical cross-check against the classical Pelgrom matching law using the mismatch coefficients embedded directly in the SKY130 model files. The Monte Carlo and Pelgrom-predicted mismatch distributions agree closely, yielding a cutoff-frequency spread of ±3.4% and a Q spread of ±5.0% about nominal, with worst-case PVT-corner shifts of up to −11.3%; supply-current extraction across the same 500-run population additionally reveals a device-to-device power-consumption spread of comparable magnitude, directly linking mismatch-driven frequency error to its power-budget cost — a fourth statistical output not reported in prior open-PDK filter studies. The resulting dataset constitutes, to the authors' knowledge, the first Monte Carlo/PVT statistical characterization of a complete Gm-C filter in an open-source PDK, cross-validated against an independent analytical model in the absence of measured silicon data. All netlists, Monte Carlo scripts, and analysis code are released publicly, providing a validated statistical baseline for SKY130 analog filter design and a reusable auditing methodology for open-source statistical device models generally.

CONFERENCES TO PUBLISH

Venue	Realistic Timing
VDAT 2027 (Intl. Symposium on VLSI Design and Test)	Deadline historically ~May, conference ~Aug; 2027 window likely opens early/mid-2027
IEEE INDICON 2027	Portal likely opens mid-2027 (2026 edition opened 22 Jun)
IEEE ISVLSI 2027	Historically opens deadline ~Feb for a summer conference
IEEE VLSID 2028	2027 edition's deadline was 27 Jul 2026, so 2028's will likely be mid-2027
IEEE iSES 2027 / INDISCON 2027	Both typically open mid-year for a Sept/Dec conference

Additional Conferences Worth Considering

Venue	Timing
IEEE ISCAS 2027 (flagship IEEE CAS Society conference)	Deadline 13 Oct 2026, notification 11 Jan 2027, conference June 2027 in Bordeaux
IEEE APCCAS (Asia Pacific Conference on Circuits and Systems)	Rotates annually across Asia-Pacific; deadlines typically follow a mid-year pattern
IEEE ICECS 2027 (flagship CAS Society conference for Region 8)	2026 edition deadline was 15 Jun 2026 (extended); 2027 likely similar mid-year deadline
IEEE AICAS (AI Circuits and Systems)	Historically ~30–40% acceptance; deadline pattern around Sept–Oct for a following-March conference

EXTENSIONS (Possible Future Work)

Six planned extensions to the current schematic-level Monte Carlo work, listed in priority order:

1. **Full-filter layout** — Draw the complete two-stage filter (not just one OTA) in Magic/KLayout, DRC/LVS clean.
2. **Post-layout Monte Carlo** — Extract real parasitics from that layout and re-run the statistics on the extracted netlist.
3. **Cross-simulator check** — Repeat the campaign in Xyce to confirm ngspice isn't giving tool-specific artifacts.
4. **Multi-topology comparison** — Run the same pipeline across 2–3 OTA topologies (diff-pair, telescopic, folded-cascode) for a designer's guideline.
5. **Calibration loop** — Add a digital bias-current trim that corrects the mismatch spread, and report before/after.
6. **One-command open release** — Package everything (schematic, layout, scripts, data) as a single reproducible, tagged repo.